

MUHAMMED SHEHEEN M T P

AI/ML ENGINEER | NLP | MACHINE TRANSLATION | LLMs | COMPUTER VISION

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PROFESSIONAL SUMMARY

AI/ML Engineer with 1 year of hands-on experience as a Paid Intern at ICFOSS, contributing to Malayalam NLP and BHASHINI language-technology workflows. Worked across linguistic data annotation, English–Malayalam corpus engineering, semantic filtering, NLLB-200 fine-tuning, OpenNMT experiments, translation evaluation, LLM application development, backend APIs, and live multilingual AI deployment. Strong in Python, PyTorch, Hugging Face Transformers, PEFT/LoRA, FastAPI, PostgreSQL, Docker, and Linux.

TECHNICAL SKILLS

Programming: Python, SQL

Machine Learning: PyTorch, TensorFlow, scikit-learn, Deep Learning, Transfer Learning, Model Evaluation

NLP / Machine Translation: Hugging Face Transformers, NLLB-200, OpenNMT, Sentence Transformers, LaBSE, tokenization, POS tagging, NER, chunking, dependency parsing, discourse annotation, morphological segmentation, summarization, paraphrasing

LLM / Generative AI: PEFT, LoRA, QLoRA, quantization, prompt engineering, Ollama, Qwen2.5-Coder

Data / Evaluation: NumPy, Pandas, Hugging Face Datasets, semantic similarity, cosine similarity, corpus filtering, BLEU, chrF, loss and perplexity analysis

Backend / Infrastructure: FastAPI, SQLAlchemy, PostgreSQL, REST APIs, Docker, Docker Compose, Judge0, Git, Linux, Jupyter, Google Colab

Computer Vision: OpenCV, YOLOv8, object detection, MediaPipe

EXPERIENCE

ICFOSS — International Centre for Free and Open Source Solutions, Government of Kerala

Paid Intern | Thiruvananthapuram, Kerala | Aug 2025 – Aug 2026

- Contributed to Malayalam NLP and machine-translation data engineering for BHASHINI, working across POS tagging, NER, chunking, dependency-parser correction, discourse annotation, morphological segmentation, summarization, and paraphrasing.
- Built and validated English–Malayalam parallel-corpus workflows using text normalization, Unicode correction, language validation, duplicate removal, length-ratio checks, and sentence-alignment review.
- Designed semantic-quality filtering with multilingual sentence embeddings and cosine similarity to identify noisy or weakly aligned bilingual sentence pairs.
- Fine-tuned Meta NLLB-200 Distilled 600M with Hugging Face Transformers and PEFT/LoRA; handled tokenization, language codes, checkpoints, adapter merging, inference, and translation evaluation with BLEU and chrF.
- Applied 8-bit quantization, gradient checkpointing, gradient accumulation, mixed precision, dynamic padding, and batch-size tuning for resource-constrained model training.
- Developed an AI Tutor backend using FastAPI, PostgreSQL, SQLAlchemy, and Docker, implementing 15 REST endpoints, adaptive learner-state logic, LLM feedback with Ollama, and secure multi-language code execution through Judge0.
- Supported live Bhashini AI deployment at the VISION 2031 conference, operating real-time ASR and multilingual translation, integrating audio input, monitoring synchronization, and troubleshooting acoustic issues.
- Prepared domain-specific Malayalam resources including dialect mappings, medicinal-plant data, OCR/ASR verification, and large-scale English–Malayalam translation batches for downstream NLP systems.

PROJECTS

AI Tutor Platform | *FastAPI, PostgreSQL, SQLAlchemy, Ollama, Judge0, Docker*

- Built a full-stack adaptive coding platform with 15 REST APIs for authentication, learner progress, problem recommendation, code evaluation, and AI feedback. Implemented adaptive prompting and learner-state logic.

English–Malayalam NMT and Corpus Quality Pipeline | *Python, PyTorch, Transformers, NLLB-200, OpenNMT, PEFT/LoRA*

- Developed workflows for bilingual corpus preparation, semantic filtering, Transformer training experiments, inference, and translation-quality evaluation; optimized NLLB fine-tuning for limited GPU resources.

Autonomous USV for Aquatic Waste Detection | *Python, YOLOv8, OpenCV, ESP32*

- Developed a computer-vision subsystem for floating-waste detection and autonomous surface-vehicle operation, integrating YOLOv8 object detection with OpenCV video processing and ESP32 control.

GestureSync | *Python, OpenCV, MediaPipe, Speech Recognition*

- Built a multimodal human-computer interaction system combining real-time hand-landmark recognition with voice-command processing for hands-free desktop control.

Linux Adaptive Tutor | *FastAPI, PostgreSQL, Ollama, Judge0*

- Built an AI-assisted Linux learning platform with adaptive lesson difficulty, multiple explanation modes, progress tracking, and secure Bash command execution.

EDUCATION

B.Tech in Information Technology — College of Engineering Thalassery, Kerala, India

Diploma in Mechanical Engineering — Swami Nithyananda Polytechnic College, Kanhangad, Kerala, India

CERTIFICATIONS

- IBM Generative AI: Prompt Engineering Basics
- IBM Generative AI: Introduction and Applications
- IBM Introduction to Artificial Intelligence
- IBM AI Foundations for Business Specialization
- IBM Python for Data Science, AI and Development
- PostgreSQL for Everybody Specialization — University of Michigan